An elastic network model with $k_{ij} = k(l_{ij})$

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1 Introduction

In this document, we pose a general elastic network model in which the parameters are the spring equilibrium lengths l_{ij}, with force constants determined by them, $k_{ij} = k(l_{ij})$. We derive its potential energy function, the potential's quadratic Taylor expansion around the minimum, its normal modes, the conformational distribution, the conformational fluctuations, the thermodynamic properties, and the activation free energy.

Once the general model is derived, given an arbitrary reference ENM, we model a mutant, by introducing perturbations to the l_{ij} of the springs connected to the mutated site.

2 The generalized elastic network model

2.1 Definition

A protein is represented by an elastic network of N nodes, which represent residues, joined in pairs by harmonic springs, which represent interactions.

Nodes

The 3D position vector of node i in a given conformation is denoted by $\mathbf{r}_i$.

The network conformation, $\mathbf{r}$, is the vector constituted by set of the node positions,

$$\mathbf{r} = (\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N) \in \mathbb{R}^{3N}. \tag{1}$$

An equilibrium conformation, a minimum of the potential defined below, is written $\mathbf{r}^e$, with node positions $\mathbf{r}_i^e$.

Pairs of nodes

Consider a pair of nodes i and j.

The vector that goes from node i to node j is given by:

$$\mathbf{r}_{ij} \equiv \mathbf{r}_j - \mathbf{r}_i \tag{2}$$

Then d_{ij}, the distance between the two nodes in conformation $\mathbf{r}$, is the norm of this vector, and $\hat{\mathbf{e}}_{ij}$ the unit vector along it, pointing from i to j:

$$d_{ij} = \|\mathbf{r}_{ij}\|, \qquad \hat{\mathbf{e}}_{ij} = \frac{\mathbf{r}_{ij}}{d_{ij}}. \tag{3}$$

$\mathbf{r}_{ij}$, d_{ij} and $\hat{\mathbf{e}}_{ij}$ are functions of the conformation and are written without arguments; their values at an equilibrium conformation carry a superscript e,

$$d^e_{ij} \equiv d_{ij}(\mathbf{r}^e), \qquad \hat{\mathbf{e}}^e_{ij} \equiv \hat{\mathbf{e}}_{ij}(\mathbf{r}^e). \tag{4}$$

Springs

Every pair of nodes is joined by a spring.

A spring connecting nodes i and j is characterised by its force constant k_{ij} and its equilibrium length l_{ij}.

We assume that the force constant of a spring is determined by its own equilibrium length through a function k,

$$k_{ij} = k(l_{ij}), \tag{5}$$

where k is a non-negative function; nothing below assumes more about it.

Potential energy

The potential energy of a network with parameters $\Lambda = \{l_{ij}\}$ at conformation $\mathbf{r}$ is a sum of harmonic terms, one per spring:

$$\boxed{V(\mathbf{r};\Lambda) = \tfrac{1}{2}\sum_{i<j} k_{ij}\big(d_{ij} - l_{ij}\big)^2.} \tag{6}$$

Every term is non-negative, so $V \geq 0$ everywhere.

2.2 Quadratic expansion of the potential energy

Let $\mathbf{r}^e$ be a minimum of the potential (6). The Taylor expansion of the potential about $\mathbf{r}^e$ gives:

$$V(\mathbf{r};\Lambda) = V(\mathbf{r}^e;\Lambda) + \sum_i \left.\frac{\partial V}{\partial \mathbf{r}_i}\right|_{\mathbf{r}^e} \cdot (\mathbf{r}_i - \mathbf{r}^e_i) + \tfrac{1}{2}\sum_{i,j} (\mathbf{r}_i - \mathbf{r}^e_i)^{\mathsf{T}} \left.\frac{\partial^2 V}{\partial \mathbf{r}_i \partial \mathbf{r}_j}\right|_{\mathbf{r}^e} (\mathbf{r}_j - \mathbf{r}^e_j) + O\big(\|\mathbf{r} - \mathbf{r}^e\|^3\big). \tag{7}$$

Writing $\mathbf{K}$ for the $3N \times 3N$ matrix of second derivatives at $\mathbf{r}^e$, the quadratic term is $\frac{1}{2}(\mathbf{r}-\mathbf{r}^e)^{\mathsf{T}}\mathbf{K}(\mathbf{r}-\mathbf{r}^e)$. The linear term vanishes, since at a minimum

$$\left.\frac{\partial V}{\partial \mathbf{r}_i}\right|_{\mathbf{r}^e} = \mathbf{0}, \qquad i = 1, \ldots, N, \tag{8}$$

and dropping the remainder leaves

$$\boxed{V(\mathbf{r};\Lambda) \simeq V(\mathbf{r}^e;\Lambda) + \tfrac{1}{2}(\mathbf{r}-\mathbf{r}^e)^{\mathsf{T}}\mathbf{K}(\mathbf{r}-\mathbf{r}^e),} \tag{9}$$

which is the quadratic approximation of the ENM's potential energy function.

Zeroth order: the minimum energy

Evaluating (6) at $\mathbf{r}^e$ gives the minimum energy

$$\boxed{V(\mathbf{r}^e;\Lambda) = \tfrac{1}{2}\sum_{i<j} k_{ij}\big(d^e_{ij} - l_{ij}\big)^2.} \tag{10}$$

Second order: the Hessian

The N^2 blocks of $\mathbf{K}$,

$$\mathbf{K}_{ij} \equiv \left.\frac{\partial^2 V}{\partial \mathbf{r}_i \partial \mathbf{r}_j}\right|_{\mathbf{r}^e}, \tag{11}$$

each 3×3, are obtained by differentiating V twice. Differentiating (6) with respect to $\mathbf{r}_i$, only the $N-1$ terms involving node i survive, and by the chain rule

$$\frac{\partial V}{\partial \mathbf{r}_i} = \sum_{j \neq i} k_{ij}\big(d_{ij} - l_{ij}\big)\frac{\partial d_{ij}}{\partial \mathbf{r}_i}. \tag{12}$$

Differentiating $d_{ij}^2 = \mathbf{r}_{ij} \cdot \mathbf{r}_{ij}$ with $\partial \mathbf{r}_{ij}/\partial \mathbf{r}_i = -\mathbf{I}$ gives

$$\frac{\partial d_{ij}}{\partial \mathbf{r}_i} = -\frac{\mathbf{r}_{ij}}{d_{ij}} \equiv -\hat{\mathbf{e}}_{ij}, \qquad \frac{\partial d_{ij}}{\partial \mathbf{r}_j} = +\hat{\mathbf{e}}_{ij}, \tag{13}$$

with $\hat{\mathbf{e}}_{ij}$ the unit vector along the spring, pointing from i to j, so

$$\frac{\partial V}{\partial \mathbf{r}_i} = -\sum_{j \neq i} k_{ij}\big(d_{ij} - l_{ij}\big)\hat{\mathbf{e}}_{ij}. \tag{14}$$

Differentiating (14) with respect to $\mathbf{r}_j$, with $j \neq i$, only the term of the sum that involves node j contributes, so

$$\mathbf{K}_{ij} = -k_{ij}\left[\frac{\partial d_{ij}}{\partial \mathbf{r}_j}\hat{\mathbf{e}}_{ij}^{\mathsf{T}} + (d_{ij} - l_{ij})\frac{\partial \hat{\mathbf{e}}_{ij}}{\partial \mathbf{r}_j}\right]_{\mathbf{r}=\mathbf{r}^e}. \tag{15}$$

The first derivative is $\partial d_{ij}/\partial \mathbf{r}_j = \hat{\mathbf{e}}_{ij}$ by (13). For the second, differentiating $\hat{\mathbf{e}}_{ij} = \mathbf{r}_{ij}/d_{ij}$ with respect to $\mathbf{r}_j$, using the product rule with $\partial \mathbf{r}_{ij}/\partial \mathbf{r}_j = \mathbf{I}$ and $\partial d_{ij}^{-1}/\partial \mathbf{r}_j = -d_{ij}^{-2}\hat{\mathbf{e}}_{ij}^{\mathsf{T}}$,

$$\frac{\partial \hat{\mathbf{e}}_{ij}}{\partial \mathbf{r}_j} = \frac{\mathbf{I}}{d_{ij}} - \frac{\mathbf{r}_{ij}\hat{\mathbf{e}}_{ij}^{\mathsf{T}}}{d_{ij}^2} = \frac{\mathbf{I} - \hat{\mathbf{e}}_{ij}\hat{\mathbf{e}}_{ij}^{\mathsf{T}}}{d_{ij}}. \tag{16}$$

Substituting both and evaluating at $\mathbf{r}^e$,

$$\mathbf{K}_{ij} = -k_{ij}\left[\hat{\mathbf{e}}_{ij}^e\big(\hat{\mathbf{e}}_{ij}^e\big)^{\mathsf{T}} + \frac{d_{ij}^e - l_{ij}}{d_{ij}^e}\Big(\mathbf{I} - \hat{\mathbf{e}}_{ij}^e\big(\hat{\mathbf{e}}_{ij}^e\big)^{\mathsf{T}}\Big)\right], \tag{17}$$

that is, defining $g_{ij} \equiv (d_{ij}^e - l_{ij})/d_{ij}^e$,

$$\boxed{\mathbf{K}_{ij} = -k_{ij}\Big[\hat{\mathbf{e}}_{ij}^e\big(\hat{\mathbf{e}}_{ij}^e\big)^{\mathsf{T}} - g_{ij}\Big(\hat{\mathbf{e}}_{ij}^e\big(\hat{\mathbf{e}}_{ij}^e\big)^{\mathsf{T}} - \mathbf{I}\Big)\Big], \qquad i \neq j.} \tag{18}$$

The diagonal blocks follow from translational invariance. Displacing every node by the same vector $\mathbf{a}$ changes no distance, so $V(\mathbf{r}_1 + \mathbf{a}, \ldots, \mathbf{r}_N + \mathbf{a}) = V(\mathbf{r}_1, \ldots, \mathbf{r}_N)$ for every conformation and every $\mathbf{a}$. Differentiating this identity twice with respect to the node positions gives $\sum_j \mathbf{K}_{ij} = \mathbf{0}$ for each i, whence

$$\boxed{\mathbf{K}_{ii} = -\sum_{j \neq i} \mathbf{K}_{ij}.} \tag{19}$$

Two properties of $\mathbf{K}$ are used below. It is symmetric, which is visible in (18): each block is a sum of symmetric matrices, and $\mathbf{K}_{ij} = \mathbf{K}_{ji}$. It is positive semi-definite, which does not follow from the equations but from $\mathbf{r}^e$ being a minimum: V cannot decrease in any direction, so $\mathbf{u}^{\mathsf{T}}\mathbf{K}\mathbf{u} \geq 0$ for every $\mathbf{u}$. Conversely, a stationary conformation at which $\mathbf{K}$ is positive semi-definite is a minimum, which is what a numerically obtained $\mathbf{r}^e$ is checked against.

2.3 Normal modes

$\mathbf{K}$ is real and symmetric, so it has $3N$ orthonormal eigenvectors $\mathbf{q}_n$ with real eigenvalues λ_n,

$$\mathbf{K}\mathbf{q}_n = \lambda_n \mathbf{q}_n, \qquad \mathbf{q}_m^{\mathsf{T}}\mathbf{q}_n = \delta_{mn}, \qquad \mathbf{K} = \sum_n \lambda_n \mathbf{q}_n \mathbf{q}_n^{\mathsf{T}}, \tag{20}$$

where δ_{mn} is 1 for $m = n$ and 0 otherwise, and $\lambda_n \geq 0$ because $\mathbf{K}$ is positive semi-definite. The eigenvectors $\mathbf{q}_n$ are the *normal modes* of the network and the eigenvalues λ_n are their force constants, carrying the same units as k_{ij}, energy per length squared.

The normal modes are an orthonormal basis of $\mathbb{R}^{3N}$, so any displacement can be written in them,

$$\mathbf{r} - \mathbf{r}^e = \sum_n u_n \mathbf{q}_n, \qquad u_n = \mathbf{q}_n^{\mathsf{T}}(\mathbf{r} - \mathbf{r}^e), \tag{21}$$

where the u_n are the coordinates of the displacement in this basis, one number per mode. In them the quadratic approximation (9) becomes

$$\boxed{V(\mathbf{r};\Lambda) \simeq V(\mathbf{r}^e;\Lambda) + \tfrac{1}{2}\sum_n \lambda_n u_n^2,} \tag{22}$$

a sum of independent terms, one per mode, with no cross terms $u_m u_n$: in this representation the modes are uncoupled, each contributing an energy set by its own force constant.

Since V depends on the conformation only through the distances (3), it is unchanged by the three translations and three rotations of the network as a whole, so those six displacements cost no energy and at least six eigenvalues vanish. Whether there are exactly six is a property of the network, not a given. A network whose Hessian has no further null directions is *infinitesimally rigid*, and then $\operatorname{rank}(\mathbf{K}) = 3N - 6$. A node held by a single spring of nonzero force constant can swing about it, and one held by two can rotate about the axis through them; each such freedom is a further null direction. In what follows the networks are assumed infinitesimally rigid, and sums and products over the $3N - 6$ nonzero eigenvalues are written $\sum_{\lambda_n \neq 0}$ and $\prod_{\lambda_n \neq 0}$.

2.4 Thermal properties

The Boltzmann distribution and the partition function

At temperature T the conformations are distributed according to

$$\rho(\mathbf{r}) = \frac{e^{-\beta V(\mathbf{r};\Lambda)}}{Z}, \qquad Z = \int e^{-\beta V(\mathbf{r};\Lambda)} d\mathbf{r}, \qquad \beta = \frac{1}{k_B T}, \tag{23}$$

the integral running over $\mathbb{R}^{3N}$. This is a configurational integral: the momenta contribute a factor that depends on the masses and on β but not on Λ, and it cancels from every difference between networks.

Inserting the quadratic approximation (9) gives a Gaussian distribution centred on $\mathbf{r}^e$,

$$\boxed{\rho(\mathbf{r}) \simeq \frac{1}{Z} e^{-\beta V(\mathbf{r}^e;\Lambda)} \exp\big(-\tfrac{\beta}{2}(\mathbf{r} - \mathbf{r}^e)^{\mathsf{T}}\mathbf{K}(\mathbf{r} - \mathbf{r}^e)\big).} \tag{24}$$

In the normal-mode representation (21) the exponent is a sum of independent terms, by (22), so the distribution factorises into one Gaussian per mode,

$$\rho \;\propto\; \prod_n e^{-\beta\lambda_n u_n^2/2}, \tag{25}$$

and so does the integral. The six modes with $\lambda_n = 0$ carry no exponent: along them ρ is flat and the integral diverges. These six directions are the translations and rotations of the network as a whole, which move it without deforming it. They are left out of Z, which is therefore the partition function of the network's internal degrees of freedom alone. Each of the remaining $3N-6$ integrals is a standard Gaussian,

$$\int_{-\infty}^{\infty} e^{-\beta\lambda_n u_n^2/2} du_n = \left(\frac{2\pi}{\beta\lambda_n}\right)^{1/2}, \tag{26}$$

so

$$\boxed{Z = e^{-\beta V(\mathbf{r}^e;\Lambda)} \prod_{\lambda_n\neq 0} \left(\frac{2\pi}{\beta\lambda_n}\right)^{1/2},} \tag{27}$$

or, in the form used below,

$$\boxed{\ln Z = -\beta V(\mathbf{r}^e;\Lambda) + \tfrac{1}{2}\sum_{\lambda_n\neq 0} \ln\frac{2\pi}{\beta\lambda_n}.} \tag{28}$$

Conformational fluctuations

The distribution (24) is a Gaussian in the displacement $\mathbf{r}-\mathbf{r}^e$, so its second moment is fixed by the matrix in the exponent. For a Gaussian $\propto \exp\big(-\frac{\beta}{2}\mathbf{x}^{\mathsf{T}}\mathbf{K}\mathbf{x}\big)$ the covariance is $(\beta\mathbf{K})^{-1}$, which requires $\mathbf{K}$ to be invertible. It is not: the six rigid-body directions have $\lambda_n = 0$, and they were left out of Z for that reason. Restricted to the complement of those directions, where the network deforms, $\mathbf{K}$ is invertible and

$$\boxed{\boldsymbol{\Sigma} = \big\langle (\mathbf{r}-\mathbf{r}^e)(\mathbf{r}-\mathbf{r}^e)^{\mathsf{T}}\big\rangle = \beta^{-1}\mathbf{K}^{+},} \tag{29}$$

with $\mathbf{K}^{+}$ the pseudo-inverse of $\mathbf{K}$: the inverse on that complement, and zero on the rigid-body directions. $\boldsymbol{\Sigma}$ is therefore the covariance of the displacements from $\mathbf{r}^e$ with no component along the rigid-body directions of $\mathbf{r}^e$: the network's fluctuations in the frame in which it neither translates nor rotates relative to $\mathbf{r}^e$. Its 3×3 blocks give the fluctuation of one node and the correlation between two,

$$\big\langle \|\mathbf{r}_i - \mathbf{r}_i^e\|^2\big\rangle = \beta^{-1}\,\mathrm{tr}\,\big(\mathbf{K}^{+}\big)_{ii}, \qquad \big\langle (\mathbf{r}_i-\mathbf{r}_i^e)(\mathbf{r}_j-\mathbf{r}_j^e)^{\mathsf{T}}\big\rangle = \beta^{-1}\big(\mathbf{K}^{+}\big)_{ij}, \tag{30}$$

the first being, up to a factor $8\pi^2/3$, a crystallographic B-factor, except that it excludes the rigid-body motion a measured B-factor includes. In Cartesian coordinates $\boldsymbol{\Sigma}$ is not diagonal: the displacements of two nodes are correlated.

In the normal-mode representation the same distribution is (25), one independent Gaussian per mode, so the coordinates u_n are uncorrelated and

$$\boxed{\langle u_m u_n\rangle = \frac{\delta_{mn}}{\beta\lambda_n}, \qquad \lambda_n \neq 0,} \tag{31}$$

each mode fluctuating with an amplitude set by its own force constant. This is $\mathbf{\Sigma}$ in the basis that diagonalises it, and it is also how the pseudo-inverse is computed,

$$\mathbf{K}^{+} = \sum_{\lambda_n \neq 0} \frac{\mathbf{q}_n \mathbf{q}_n^{\mathsf{T}}}{\lambda_n}. \tag{32}$$

Internal energy, entropy, free energy

The internal energy is the mean potential energy, $U = \langle V \rangle$, obtained from the partition function as

$$U = -\frac{\partial \ln Z}{\partial \beta}, \tag{33}$$

which by (28) gives

$$\boxed{U = V(\mathbf{r}^e; \Lambda) + \frac{3N-6}{2\beta},} \tag{34}$$

the minimum energy plus $\frac{1}{2\beta}$ per non-rigid degree of freedom.

The entropy is $S = -k_B \langle \ln \rho \rangle$, and by (23) $\ln \rho = -\beta V - \ln Z$, so

$$S = k_B\big(\beta \langle V \rangle + \ln Z\big) = k_B\big(\beta U + \ln Z\big). \tag{35}$$

Substituting (34) and (28), the terms in $V(\mathbf{r}^e; \Lambda)$ cancel and

$$TS = \frac{1}{\beta}\big(\beta U + \ln Z\big) = \frac{3N-6}{2\beta} + \frac{1}{2\beta} \sum_{\lambda_n \neq 0} \ln \frac{2\pi}{\beta \lambda_n}. \tag{36}$$

The sum has $3N-6$ terms, so the first term is $\frac{1}{2\beta} \sum_{\lambda_n \neq 0} 1$ and the two can be written as one sum,

$$\boxed{TS = \frac{1}{2\beta} \sum_{\lambda_n \neq 0} \left[\ln \frac{2\pi}{\beta \lambda_n} + 1 \right].} \tag{37}$$

The Helmholtz free energy is $A = U - TS$, which by (34) and (37) is

$$\boxed{A = V(\mathbf{r}^e; \Lambda) - \frac{1}{2\beta} \sum_{\lambda_n \neq 0} \ln \frac{2\pi}{\beta \lambda_n}.} \tag{38}$$

Equivalently $A = -\beta^{-1} \ln Z$, by (28).

The argument of each logarithm is $2\pi/\beta\lambda_n$, which has units of length squared, since λ_n is an energy per length squared. The numerical values of TS and A therefore depend on the unit of length chosen, and A may fall below $V(\mathbf{r}^e; \Lambda)$; the same applies to the momentum factor and the rigid-body directions left out of Z. Differences between two networks at the same temperature are free of all three, as are comparisons at a fixed choice of units.

2.5 Activation free energy

Active site and transition state

Let $\mathcal{A}$ be a set of $n_{\mathcal{A}}$ nodes, the active site, and

$$\mathbf{x} = (\mathbf{r}_i)_{i\in\mathcal{A}} \in \mathbb{R}^{3n_{\mathcal{A}}} \tag{39}$$

their positions, with equilibrium value $\mathbf{x}^e$. The transition state is the network with the active site held at a given conformation $\mathbf{x}^\ddagger$, expressed in the frame of $\mathbf{r}^e$, and every other node free. Its partition function is

$$Z^\ddagger = \int e^{-\beta V(\mathbf{r};\Lambda)}\,\delta\big(\mathbf{x}-\mathbf{x}^\ddagger\big)\,d\mathbf{r}, \tag{40}$$

over the same internal degrees of freedom as Z, and the activation free energy is

$$\Delta A^\ddagger \equiv A^\ddagger - A = -\beta^{-1}\ln\frac{Z^\ddagger}{Z}. \tag{41}$$

The marginal distribution of the active site

By (23),

$$\frac{Z^\ddagger}{Z} = \int \rho(\mathbf{r})\,\delta\big(\mathbf{x}-\mathbf{x}^\ddagger\big)\,d\mathbf{r} = \rho_{\mathcal{A}}(\mathbf{x}^\ddagger), \tag{42}$$

with $\rho_{\mathcal{A}}$ the marginal density of the active-site coordinates. In the quadratic approximation ρ is the Gaussian (24), with mean $\mathbf{r}^e$ and covariance (29), and its marginal is the Gaussian with mean $\mathbf{x}^e$ and covariance the $\mathcal{A}\mathcal{A}$ block, $\mathbf{\Sigma}_{\mathcal{A}\mathcal{A}} = \beta^{-1}(\mathbf{K}^+)_{\mathcal{A}\mathcal{A}}$. Define the effective Hessian of the active site,

$$\boxed{\mathbf{K}_{\mathcal{A}} \equiv \Big[\big(\mathbf{K}^+\big)_{\mathcal{A}\mathcal{A}}\Big]^{-1},} \tag{43}$$

a $3n_{\mathcal{A}}\times 3n_{\mathcal{A}}$ matrix with eigenvalues κ_a, $a=1,\dots,3n_{\mathcal{A}}$. $\mathbf{K}^+$ is positive semi-definite with null space the rigid-body directions, so $(\mathbf{K}^+)_{\mathcal{A}\mathcal{A}}$ is singular only if some rigid-body displacement vanishes on every node outside $\mathcal{A}$, which requires those nodes to be collinear; otherwise the inverse exists and the κ_a are positive. Then

$$\rho_{\mathcal{A}}(\mathbf{x}) = \prod_a \left(\frac{\beta\kappa_a}{2\pi}\right)^{1/2} \exp\Big(-\tfrac{\beta}{2}\big(\mathbf{x}-\mathbf{x}^e\big)^{\mathsf{T}}\mathbf{K}_{\mathcal{A}}\big(\mathbf{x}-\mathbf{x}^e\big)\Big). \tag{44}$$

The activation free energy

Substituting (44) into (41),

$$\boxed{\Delta A^\ddagger = \tfrac{1}{2}\big(\mathbf{x}^\ddagger-\mathbf{x}^e\big)^{\mathsf{T}}\mathbf{K}_{\mathcal{A}}\big(\mathbf{x}^\ddagger-\mathbf{x}^e\big) + \frac{1}{2\beta}\sum_a \ln\frac{2\pi}{\beta\kappa_a}.} \tag{45}$$

The first term,

$$\Delta V^\ddagger \equiv \tfrac{1}{2}\big(\mathbf{x}^\ddagger-\mathbf{x}^e\big)^{\mathsf{T}}\mathbf{K}_{\mathcal{A}}\big(\mathbf{x}^\ddagger-\mathbf{x}^e\big), \tag{46}$$

is the minimum of (9) over the conformations with $\mathbf{x}=\mathbf{x}^\ddagger$, less $V(\mathbf{r}^e;\Lambda)$. In the transition state the $3n_{\mathcal{A}}$ coordinates of the active site are fixed and the remaining $3N-6-3n_{\mathcal{A}}$ internal degrees of

freedom fluctuate harmonically about that constrained minimum, so, as in (34), $U^\ddagger = V(\mathbf{r}^e;\Lambda) + \Delta V^\ddagger + (3N-6-3n_{\mathcal{A}})/2\beta$, and

$$\Delta U^\ddagger = \Delta V^\ddagger - \frac{3n_{\mathcal{A}}}{2\beta}, \qquad -T\Delta S^\ddagger = \frac{1}{2\beta}\sum_a \left[\ln\frac{2\pi}{\beta\kappa_a} + 1\right], \tag{47}$$

the second being T times the entropy of $\rho_{\mathcal{A}}$, which the transition state loses. The transition state has $3n_{\mathcal{A}}$ fewer degrees of freedom than the network, so, unlike the differences of Section 3.2, $\Delta A^\ddagger$ and $T\Delta S^\ddagger$ depend on the unit of length; their differences between two networks do not.

3 Mutations

3.1 Reference and mutant

The reference network

Everything above applies to any network. We now read it as describing a particular one, the *reference*, with parameters $\Lambda = \{l_{ij}\}$, force constants $k_{ij} = k(l_{ij})$, potential $V(\mathbf{r};\Lambda)$, equilibrium conformation $\mathbf{r}^e$, equilibrium distances d^e_{ij}, Hessian $\mathbf{K}$, spectrum $\{\lambda_n\}$ and thermodynamic properties U, S, A. The mutant network is written with a prime on each of these: Λ', k'_{ij}, V', $\mathbf{r}^{e\prime}$, $d^{e\prime}_{ij}$, $\mathbf{K}'$, λ'_n, U', S', A'.

Mutation

We model a mutation at site m as a perturbation of the equilibrium lengths of the springs that connect m to the rest of the network, leaving all others unchanged:

$$l'_{ij} = l_{ij} + \delta l_{ij}, \qquad \delta l_{ij} = 0 \ \text{ unless } i = m \text{ or } j = m. \tag{48}$$

The δl_{ij} are arbitrary: nothing below assumes them small, random, or related to one another.

The mutant potential is (6) evaluated with these parameters,

$$V'(\mathbf{r}) \equiv V(\mathbf{r};\Lambda') = \tfrac{1}{2}\sum_{i<j} k\big(l_{ij}+\delta l_{ij}\big)\big(d_{ij}-l_{ij}-\delta l_{ij}\big)^2. \tag{49}$$

The mutation changes both the equilibrium lengths and, through (5), the force constants. Define

$$\delta k_{ij} \equiv k\big(l_{ij}+\delta l_{ij}\big) - k\big(l_{ij}\big), \qquad \text{so} \qquad k'_{ij} = k_{ij} + \delta k_{ij}. \tag{50}$$

The mutant has its own equilibrium conformation $\mathbf{r}^{e\prime}$, obtained by minimising (49), its own equilibrium distances $d^{e\prime}_{ij} \equiv d_{ij}(\mathbf{r}^{e\prime})$, its own Hessian $\mathbf{K}'$ and spectrum $\{\lambda'_n\}$, and its own U', S', A'. Note that $\mathbf{r}^{e\prime}$ is obtained by minimising (49) itself. Its first-order approximation is the linear response $\mathbf{r}^{e\prime} - \mathbf{r}^e \simeq \mathbf{K}^+\mathbf{f}$, with $\mathbf{f}$ the force that the change of parameters exerts at $\mathbf{r}^e$ and $\mathbf{K}^+$ the pseudo-inverse (32); this is what the linearly forced elastic network model computes, and it is not used below.

Even if the reference is unfrustrated, the mutant will in general be frustrated: no conformation need satisfy $d_{ij} = l'_{ij}$ for every spring at once, so the mutant settles at a compromise with $d^{e\prime}_{ij} \neq l'_{ij}$ (Appendix B).

3.2 Mutational changes

Changes in the thermodynamic properties

The mutant's properties are those of Section 2.4 evaluated with its own parameters, and the changes are their differences. Both require the two networks to have the same number of non-null directions. A point mutation neither adds nor removes a node, so both Hessians are $3N \times 3N$, and if both networks are infinitesimally rigid both have $3N - 6$ nonzero eigenvalues.

Write $\Delta V_{\min}$ for the difference between the two minimum energies,

$$\Delta V_{\min} \equiv V'(\mathbf{r}^{e\prime}) - V(\mathbf{r}^{e}). \tag{51}$$

By (34) the internal energies are $U = V(\mathbf{r}^e; \Lambda) + (3N-6)/2\beta$ and $U' = V'(\mathbf{r}^{e\prime}) + (3N-6)/2\beta$, whose second terms are equal, so

$$\boxed{\Delta U = \Delta V_{\min}.} \tag{52}$$

For the entropy, separate in (37) the terms that depend on the network from those that do not,

$$TS = \frac{1}{2\beta}\left[(3N-6)\left(\ln\frac{2\pi}{\beta}+1\right) - \sum_{\lambda_n\neq 0}\ln\lambda_n\right], \tag{53}$$

and likewise for the mutant with λ'_n. The first term is the same for both, so on subtracting,

$$T\Delta S = \frac{1}{2\beta}\left[\sum_{\lambda_n\neq 0}\ln\lambda_n - \sum_{\lambda'_n\neq 0}\ln\lambda'_n\right]. \tag{54}$$

Writing the sums of logarithms as logarithms of products,

$$\boxed{T\Delta S = \frac{1}{2\beta}\ln\frac{\prod_{\lambda_n\neq 0}\lambda_n}{\prod_{\lambda'_n\neq 0}\lambda'_n}.} \tag{55}$$

The two spectra enter only through these products; no correspondence between the modes of the reference and those of the mutant is used.

The free energy change is $\Delta A = \Delta U - T\Delta S$, which by (52) and (55) is

$$\boxed{\Delta A = \Delta V_{\min} - \frac{1}{2\beta}\ln\frac{\prod_{\lambda_n\neq 0}\lambda_n}{\prod_{\lambda'_n\neq 0}\lambda'_n}.} \tag{56}$$

Both (55) and (56) depend on the spectra only through the ratio of the two products, so the choice of length unit does not affect them. In the protein literature (56) is what is meant by $\Delta\Delta G$ for the folded state.

A mutation therefore changes the free energy in two ways: it shifts the minimum of the potential, and it changes the spectrum of $\mathbf{K}$, which by (18) depends on the strains through g_{ij} and on the force constants k_{ij}. A mutation that leaves the minimum energy untouched can still change the free energy, through the spectrum alone.

The change in the minimum energy

$\Delta V_{\min}$ can be computed in two stages, by changing the parameters first and letting the network relax afterwards. The first stage is the cost of changing the parameters at fixed conformation,

$$\Delta V_{\text{stress}} = V'(\mathbf{r}^e) - V(\mathbf{r}^e) = \tfrac{1}{2}\sum_{i<j}\Big[k'_{ij}\big(d^e_{ij} - l'_{ij}\big)^2 - k_{ij}\big(d^e_{ij} - l_{ij}\big)^2\Big], \tag{57}$$

which needs only the reference structure and the two parameter sets. For an unfrustrated reference $d^e_{ij} = l_{ij}$, and it reduces to $\frac{1}{2}\sum_{i<j} k'_{ij}\delta l^2_{ij} \geq 0$: quadratic in the perturbation, and positive whichever way each equilibrium length moves. A frustrated reference keeps the terms linear in δl_{ij}, of either sign, so its minimum energy responds to a mutation at first order in the perturbation while an unfrustrated one does not. the second the energy released as the mutant relaxes from $\mathbf{r}^e$ to its own equilibrium,

$$\Delta V_{\text{relax}} = V'(\mathbf{r}^{e\prime}) - V'(\mathbf{r}^e), \tag{58}$$

and since $V'(\mathbf{r}^e)$ cancels between them,

$$\boxed{\Delta V_{\min} = \Delta V_{\text{stress}} + \Delta V_{\text{relax}}.} \tag{59}$$

Since $\mathbf{r}^{e\prime}$ minimises V', we have $V'(\mathbf{r}^{e\prime}) \leq V'(\mathbf{r})$ for every $\mathbf{r}$, and in particular for $\mathbf{r} = \mathbf{r}^e$, so

$$\Delta V_{\text{relax}} \leq 0. \tag{60}$$

The change in the activation free energy

The mutant's activation free energy $\Delta A^{\ddagger\prime}$ is (45) for the same active site and the same $\mathbf{x}^\ddagger$, now in the frame of $\mathbf{r}^{e\prime}$, with the mutant's own $\mathbf{x}^{e\prime}$ and $\mathbf{K}'_{\mathcal{A}} \equiv [(\mathbf{K}'^{+})_{\mathcal{A}\mathcal{A}}]^{-1}$, of eigenvalues κ'_a. The terms $3n_{\mathcal{A}}/2\beta$ of (47) are the same for both networks, so $\Delta\Delta U^\ddagger = \Delta\Delta V^\ddagger$, with

$$\Delta\Delta V^\ddagger \equiv \tfrac{1}{2}\big(\mathbf{x}^\ddagger - \mathbf{x}^{e\prime}\big)^{\mathsf{T}}\mathbf{K}'_{\mathcal{A}}\big(\mathbf{x}^\ddagger - \mathbf{x}^{e\prime}\big) - \tfrac{1}{2}\big(\mathbf{x}^\ddagger - \mathbf{x}^{e}\big)^{\mathsf{T}}\mathbf{K}_{\mathcal{A}}\big(\mathbf{x}^\ddagger - \mathbf{x}^{e}\big), \tag{61}$$

and

$$\boxed{T\Delta\Delta S^\ddagger = \frac{1}{2\beta}\ln\frac{\prod_a \kappa'_a}{\prod_a \kappa_a}, \qquad \Delta\Delta A^\ddagger = \Delta\Delta V^\ddagger - \frac{1}{2\beta}\ln\frac{\prod_a \kappa'_a}{\prod_a \kappa_a}.} \tag{62}$$

When the reference active site is at the transition-state conformation, $\mathbf{x}^\ddagger = \mathbf{x}^e$, the reference term of (61) vanishes and $\Delta\Delta V^\ddagger = \frac{1}{2}(\mathbf{x}^e - \mathbf{x}^{e\prime})^{\mathsf{T}}\mathbf{K}'_{\mathcal{A}}(\mathbf{x}^e - \mathbf{x}^{e\prime}) \geq 0$.

Appendices

A Open questions and work to come

The following are not settled here.

- The quadratic truncation (9) is the only approximation in the document. Its accuracy for a frustrated network has not been examined.
- How large $|d_{ij}^e - l_{ij}|$ may become before a compressed spring ($d_{ij}^e < l_{ij}$), with its negative transverse force constant (65), ceases to represent a residue interaction.
- A mutation can cost the network its infinitesimal rigidity: if k vanishes over some range of lengths and enough springs of the mutated site are carried into it, the site acquires a rotational freedom. The mutant Hessian then has more than six null directions, the counts in Section 3.2 no longer match, and neither (52) nor (55) holds as written.
- How the parameters Λ should be chosen. Setting $l_{ij} = d_{ij}(\mathbf{r}^*)$ for a chosen structure $\mathbf{r}^*$ yields an unfrustrated network, but nothing selects it, and whether a frustrated network is the better model of a native structure — and if so, which one — is open.
- How the transition state is placed. $\Delta A^\ddagger$ depends on the position of $\mathbf{x}^\ddagger$ relative to $\mathbf{r}^e$, not only on its shape, because the distribution (24) has no rigid-body freedom. Whether $\mathbf{x}^\ddagger$ should be superposed onto $\mathbf{r}^e$, and on which nodes, or integrated over its rigid-body placements, is not settled.

B Frustration

Realisable parameters

The parameters Λ are *realisable* when some conformation has every spring at its own equilibrium length,

$$d_{ij}(\mathbf{r}^*) = l_{ij} \quad \text{for every spring with } k_{ij} \neq 0, \tag{63}$$

for at least one $\mathbf{r}^* \in \mathbb{R}^{3N}$. This is a strong requirement. The network has $\frac{1}{2}N(N-1)$ springs but only $3N-6$ independent coordinates, so for $N > 4$ the equilibrium lengths far outnumber the freedoms available to satisfy them: they cannot be chosen independently and still correspond to a set of points in three dimensions. Reading them off a structure, $l_{ij} = d_{ij}(\mathbf{r}^*)$, makes them realisable by construction, which is what a standard elastic network model does. Choosing them in any other way does not.

When Λ is realisable, $\mathbf{r}^*$ gives $V(\mathbf{r}^*;\Lambda) = 0$ by (6), which is the least value V can take, so $\mathbf{r}^*$ is a global minimum and the network is *unfrustrated*. Otherwise no conformation relaxes every spring at once, and the network is *frustrated*: at its minimum

$$d_{ij}^e \neq l_{ij} \quad \text{for at least one spring with } k_{ij} \neq 0. \tag{64}$$

Such a spring is strained at equilibrium, stretched if $d_{ij}^e > l_{ij}$ and compressed if $d_{ij}^e < l_{ij}$. The equilibrium condition is (14) set to zero: the forces on each node cancel, which does not require the force exerted by each spring to vanish.

The minimum energy

By (10) the minimum energy is a sum of non-negative terms, one per spring, each vanishing only for a relaxed spring. A frustrated network therefore has $V(\mathbf{r}^e;\Lambda) > 0$: it stores elastic energy even in its own equilibrium conformation, whereas an unfrustrated one has $V(\mathbf{r}^e;\Lambda) = 0$.

Longitudinal and transverse stiffness

Equation (18) contains two projectors. Writing $\mathbf{P}_\perp = \mathbf{I} - \hat{\mathbf{e}}^e_{ij}\big(\hat{\mathbf{e}}^e_{ij}\big)^{\mathsf{T}}$ for the projector onto the plane perpendicular to the spring, it may be rearranged as

$$\mathbf{K}_{ij} = -k_{ij}\Big[\hat{\mathbf{e}}^e_{ij}\big(\hat{\mathbf{e}}^e_{ij}\big)^{\mathsf{T}} + g_{ij}\mathbf{P}_\perp\Big], \tag{65}$$

so a spring carries two force constants rather than one: a longitudinal k_{ij} resisting change of the distance, and a transverse $k_{ij}g_{ij}$, doubly degenerate, resisting rotation of the spring at fixed distance.

The transverse force constant is proportional to $g_{ij} = (d^e_{ij} - l_{ij})/d^e_{ij}$, and so vanishes for a relaxed spring and changes sign with the strain. A stretched spring resists transverse displacement, as a taut string does. A compressed one has a negative transverse force constant: displacing the two nodes sideways increases their separation, which a compressed spring favours. Note that it is the transverse force constant that is negative, not k_{ij}, which is non-negative by (5).

What frustration changes

For an unfrustrated network $g_{ij} = 0$ for every spring, the transverse term of (65) is absent, and each off-diagonal block reduces to $-k_{ij}\hat{\mathbf{e}}^e_{ij}\big(\hat{\mathbf{e}}^e_{ij}\big)^{\mathsf{T}}$, a matrix of rank one. Frustration is what makes the transverse term present, and it enters $\mathbf{K}$ through g_{ij} alone.

Since $\mathbf{K}$ differs, so do its eigenvalues and eigenvectors, and hence the normal modes of Section 2.3, the partition function and thermodynamic functions of Section 2.4, and the fluctuations of Section 2.4. None of those derivations assumes an unfrustrated network, so all of them hold as written; what changes is the matrix they are applied to.